

COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2021 – Paper 7

3 Economics, Law and Ethics (ah793)

- Principles of law (a) You incorporate existing open source code into a program you are writing. What intellectual property aspects will you need to consider? [4 marks]

Answer: This is a copyright matter (1 mark). You will need to consider the type of licence the code has been made available under (1 mark). For example, if it is a GPL licence, your software will need to also be open source (1 mark). Creative commons licences will have author-specified conditions, such as if the code can be used for commercial purposes (1 mark).

- Market failure;
Principles of law (b) What are the economic arguments for and against making code you write open source? [6 marks]

Answer: The argument for making your code open source is that it becomes a public good. This is a net positive, as everyone has the potential to benefit from your contribution. This can also contribute to generating network effects, including a community where others contribute to maintaining and extending your code (1 mark). You may still economically benefit from your work, for example, you may charge for support (1 mark). While your code may be used by others, you can specify the terms under which this is permitted (1 mark).

The argument against making your code open source is that may be more difficult to attract funding (1 mark). For example, venture capitalists like to see defensible intellectual property (1 mark). You may also be bound by contract, such as to your employer (1 mark).

- Philosophies of ethics (c) You are approached by someone who claims to have identified a vulnerability in your popular software application. They demand you make them a payment, otherwise they will release the vulnerability publicly. The vulnerability is fixable and has not (yet) been exploited, however it would likely generate adverse publicity. What is the ethical approach to take? [5 marks]

Answer: You should not pay blackmailers, as it incentivises them to continue. You should fix the vulnerability as soon as possible, and ensure that the remediation efforts are taken up (1 mark). You may want to do this by releasing patches and requesting your users to install these (1 mark). Ideally, you would already have a bug bounty program set up, but If you do not you may want to establish one (1 mark). This will enable you to incentivise responsible vulnerability disclosure, and enable those who do so to be rewarded financially (1 mark). Responsible vulnerability disclosure, which you want to encourage, includes making a confidential disclosure, and then disclosing the vulnerability publicly if it is not fixed after a period of time (1 mark).

- Law and the Internet (d) You are approached by someone who claims to have identified a vulnerability in your corporate servers and breached your customer database. They demand you make them a payment, otherwise they will release the personal information. Your logs confirm the database has been breached. What is the lawful approach to take? [5 marks]

Answer: You have a legal requirement, in accordance with the GDPR (1 mark), to report the data breach to the regulator (in general, in the UK this is the ICO, although some sectors, such as critical infrastructure, may also need to report to other regulators) within 72 hours

— *Solution notes* —

(2 marks). If the data are high risk (e.g. includes payment information) you also have a requirement to notify the data subjects (2 marks).
